DR. TIM'S TUTORIALS

JACKPOT CHEMISTRY

Questions with Answers Below Each Question

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CHM 102/112 — 2023/2024 Academic Session

1. Calculate the oxidation number of chromium in CrO2■ (A) +6 (B) +1 (C) +3 (D) +5

Answer: Option B — +1

2. The term "Organic chemistry" was first coined in what year? (A) Jons Jacob Berzelius in 1807 (B) Friedrich Wohler in 1828 (C) Lavoisier in 1815 (D) Kekulé in 1921

Answer: Option A — 1807

3. The doctrine of vitalism believed that living systems possessed: (A) Mean force (B) Vital force (C) Vital energy (D) Vital power

Answer: Option B — Vital Force

4. Carbon can form a vast number of compounds because it can: (A) Donate electrons (B) Gain electrons (C) Catenate (D) Form ionic bonds

Answer: Option C — Catenate

5. What is the process of mixing orbitals of different energies to give a set of orbitals of equivalent energy called? (A) Orbital mixing (B) Equivalent energy (C) Specialization (D) Hybridization

Answer: Option D — Hybridization

6. Oxide of carbon formed when carbon burns with limited supply of oxygen is called _____ (A) C3O2 (B) CO2 (C) H2CO3 (D) CO

Answer: Option D — CO

7. The common soda-lime glass is made from (A) Na■CO■ and SiO■ (B) K■CO■ and SiO■ (C) CaCO■ and SiO■ (D) Li■CO■ and SiO■

Answer: Option A — Na■CO■ and SiO■

8. The electrical charge an element in a molecule or ion appears to have as determined by set rules is known as _____ (A) Charge (B) Oxidation number (C) Ion (D) Radical

Answer: Option B — Oxidation number

9. Chlorination of alkanes is an example of free radical _____ reaction. (A) Rearrangement (B) Elimination (C) Substitution (D) Addition

Answer: Option C — Substitution

10. What prefix is used when three carbon atoms are removed from the parent alkane compound? (A) Triseco (B) Trisnor (C) Trisdes (D) Triscyclo

Answer: Option C — Trisnor

11. Which hydrocarbon would be expected from the reaction of n-butyl chloride with sodium metal? (A) Octane (B) Decane (C) Nonane (D) Pentane

Answer: Option A — Octane

12. Which compound is most acidic? (A) Alkane (B) Alkene (C) Alkyne (D) All are equally acidic

Answer: Option C — Alkyne

13. When an enantiomer rotates a plane-polarized light in a clockwise direction, it is said to be _____ (A) Levorotatory (B) Dextrorotatory (C) Racemate (D) None

Answer: Option B — Dextrorotatory

14. Two diastereomers that differ at only one stereocenter are referred to as _____. (A) Epimers (B) Enantiomers (C) Regioners (D) Anomers

Answer: Option A — Epimers

15. One of these is not a type of constitutional isomerism? (A) Chain isomerism (B) Geometrical isomerism (C) Position isomerism (D) Functional isomerism

Answer: Option B — Geometric Isomerism (Cis and trans)

16. Formation of racemic mixture is as a result of ____ (A) External compensation (B) Internal compensation (C) Isomerisation (D) Non compensation

Answer: Option D — Non-compensation

17. How many stereoisomers are possible in H■CCH(OH)CH(OH)COOH? (A) 2 (B) 3 (C) 4 (D) 5

Answer: Option C — 4

18. A unit cell having the smallest possible volume and lattice point at each corner is called _____ (A) Body-centered cell (B) Primitive cell (C) Based-centered cell (D) Face-centered cell

Answer: Option B — Primitive cell

19. On the basis of morphology, the two types of solids are _____ (A) Crystalline and amorphous (B) Ionic and Covalent (C) Network and hydrogen bonded (D) Metallic and semiconductor

Answer: Option A — Crystalline and Amorphous

20. The chemistry of the solid state deals with the following except _____ (A) Synthesis of solids (B) Structure of solid (C) Properties and applications of solids (D) Destruction of solids

Answer: Option D — Destruction of solids

21. Which of the following is *not* a class of solid on the basis of bonding type? (A) Ionic solid (B) Metallic solid (C) Covalent solid (D) Amorphous solid

Answer: Option D — Amorphous solid.

22. The oxidation products of metals are generally ___ in size than their reactants. (A) Bigger (B) Smaller (C) None of the above (D) All of the above

Answer: Option A — Bigger

23. When a piece of K is dropped in water, two things happen: (A) Hydrogen evolves and water becomes alkaline (B) Gas evolves and water is formed (C) Effervescence evolves and water is left (D) All of the above

Answer: Option A — Hydrogen evolves and water becomes alkaline

24. Metals are Oxidants because _____ and non-metals are reductants because____ respectively. (A) Loss and gain of electrons (B) Loss and loss of electrons (C) Gain and gain of electrons (D) Gain and loss of electrons

Answer: Option A — Loss and gain of electrons.

25. Elements in S-block are highly electropositive because _____. (A) They are very reactive solids. (B) They are nonmetals that can give out 1 or 2 electrons (C) They can form cations by accepting 1 or 2 electrons (D) They are metals that can give out 1 or 2 electrons

Answer: Option D — They are metals that can give out 1 or 2 electrons

26. Which of the following alcohols is used as an anti-freezing agent? (A) Propan-1,2,3-triol (B) Propylene Glycol (C) Glycerol (D) Ethylene Glycol

Answer: Option D — Ethylene Glycol

27. The reaction of ethanol with calcium hydroxide produces (A) Calcium ethoxide and water (B) Calcium ethoxide and hydrogen (C) Calcium ethoxide and oxygen (D) No reaction

Answer: Option A — Calcium ethoxide and water.

28. The gas evolved when Propanoic acid reacts with sodium hydrogen trioxocarbonate (iv) (NaHCO■) is__. (A) Oxygen (B) Propane (C) Methane (D) Carbon dioxide

Answer: Option D — Carbon dioxide.

29. The decolourization of the purple colour of KMnO4 when ethene gas is made to pass through the compound is _____ (A) Ethylene Glycol (B) Ethanol (C) Ethanoic Acid (D) Ethyne

Answer: Option A — Ethylene Glycol

30. The IUPAC name for the compound CH■CH(CH■)CH■CH=CHCH■CHO is _____ (A) 2-methylhept-4-enal (B) 6-methylhept-3-enal (C) 6-methylhept-4-enal (D) 2-methylhept-4-enal

Answer: Option B — 6-methylhept-3-enal.

31. Which of the following are steps involved in the identification of unknown compounds except? (A) Ignition test (B) Determination of melting point (C) Esterification (D) Solubility behavior in different solvents

Answer: Option C — Esterification

32. _____ is a bittering agent that is put into house hold substances like toilet cleaner to stop children from eating them? (A) Denatonium benzoate (B) Decantation (C) Permanganate (D) Calcium ethoxide

Answer: Option A — Denatonium benzoate

33. Component parts of volatile liquid mixture can be separated by a technique referred to as (A) Fractional Distillation (B) Extraction (C) Crystallization (D) Evaporation

Answer: Option A — Fractional distillation

34. In quantitative analysis of organic compounds, the covalently bonded halogens must be changed into halide ions by _____ (A) Carius method (B) Kjeldahl method (C) Malk's method (D) X-ray method

Answer: Option A — Carius method

35. Isomers that differ in spatial arrangement of atoms rather than the order of atomic connectivity are called _____ (A) Constitutional isomers (B) Skeletal Isomers (C) Position Isomers (D) Stereoisomers

Answer: Option D — Stereoisomers.

36. What is the polytrade name for chlorothene polymer _____ (A) Polypropylene (B) Polystyrene (C) Polyvinyl chloride (D) Polythene

Answer: Option C — Polyvinyl chloride (PVC).

37. The following are types of unit cell except _____ (A) Primitive cell (B) Body centered cell (C) Face centered cell (D) Triclinic cell

Answer: Option D — Triclinic cell

38. Which of the following best describes a cubic crystal system? (A) a = b = c; $\alpha = \beta = \gamma = 90°$ (B) a = b $\neq$ c; $\alpha = \beta = \gamma = 90°$ (C) a $\neq$ b $\neq$ c; $\alpha = \beta = \gamma = 90°$ (D) a = b = c; $\alpha = \beta = \gamma \neq 90°$

Answer: Option A — a = b = c; $\alpha = \beta = \gamma = 90°$.

39. A face-centered orthorhombic crystal has _____ number of unit cells (A) 1 (B) 2 (C) 3 (D) 4

Answer: Option D — 4.

40. A combination of the seven crystal systems and the four types of lattice types of lattice gives ____ number of Bravais lattices of a structure. (A) 7 (B) 10 (C) 14 (D) 4

Answer: Option C — 14.

41. What is the IUPAC name for the compound, CH3CH2CH2NHCH3 (A) Propylethylamine (B) Methylpropylamine (C) Ethylpropylamine (D) Ethylpropylamide

Answer: Option B — N-methylpropylamine.

42. What is the common name for primary aromatic amine whose methyl group is at the parade position? (A) p-aniline (B) p-toluene (C) p-toluidine (D) p-toluic acid

Answer: Option C — p-toluidine

43. Phenyl is more acidic than alcohol due to (A) Aromaticity (B) Resonance stabilization (C) Nucleophilicity (D) Resonance de-stabilization.

Answer: Option B — Resonance Stability

44. What is the function of H■SO■ in the reaction of alcohol and carboxylic acid? (A) Hydrating agent (B) De-hydrating agent (C) Solvating (D) Catalyst

Answer: Option B — De-hydrating agent

45. The reagent used in the laboratory to identify carbonyl compounds is _____ (A) Benedict reagent (B) Fehling's reagent (C) Lucas reagent (D) 2,4-dinitrophenylhydrazine

Answer: Option D — 2,4-DNP.

46. In which of these compounds does nitrogen exhibit the oxidation state of +5? (A) N■O (B) NH■Cl (C) N■O■ (D) NO

Answer: Option C — N■O■.

47. Elements in p-block that are generally referred to as salt formers are (A) Alkali metals (B) Halogens (C) Alkali earth metals (D) Noble gas

Answer: Option B — Halogens

48. Given the common endings of the names of the binary compound of groups IV, V and VI respectively. (A) -ane, -ide, -ine (B) -ide, -ene, -ide (C) -ane, -ide, -one (D) All of the above.

Answer: Option B — -ide, -ene, -ide

49. In the reaction N■ + 3H■ → 2NH■ (Haber's process), Fe is used as what? (A) A metal (B) An oxidant (C) A reactant (D) A catalyst

Answer: Option D — A catalyst.

50. Which of the following stereoisomers exists as a meso form? (A) HOOCCH(OH)CH(OH)COOH (B) HOOCCH(OH)CH(OH)CH3 (C) HOOCCH(CH3)CH(OH)COOH (D) H3CCH(OH)CH(OH)COOH

Answer: Option A — HOOCCH(OH)CH(OH)COOH (Tartaric acid).

51. Molecules that lack a plane of symmetry are said to be _____. Molecules that lack a plane of symmetry are said to be.

Answer: ANSWER — CHIRAL

52. In the given redox reaction: 2K■Cr■O■(g) + H■O(l) + 3S(g) → 3SO■(g) + 4KOH(aq) + 2Cr■O■(s) which substance is the reducing agent? (A) S (B) K■Cr■O■ (C) SO■ (D) H■O

Answer: Option A — S (Sulfur)

53. In a redox reaction, the atom which undergoes reduction is called (A) Reducing agent (B) Reduction (C) Oxidation (D) Oxidizing agent

Answer: Option D — Oxidizing agent or Oxidant.

54. The following are natural products with alkene functionality EXCEPT? (A) Vitamin A (B) Cholestane (C) Carvone (Spearmint oil) (D) Limonene (Citrus fruit)

Answer: Option B — Cholestane

55. A group obtained by removing a hydrogen atom from alkene is called (A) Alkenyl (B) Alkynyl (C) Alkyl (D) Alryl

Answer: Option A — Alkenyl.

56. The common/old names for Alkanes, Alkenes, and Alkynes are ____ respectively. (A) Paraffins, Olefins, and Acetylenes (B) Paraffins, Acetylenes, and Olefins (C) Olefins, Paraffins, and Acetylenes (D) Acetylenes, Olefins, and Paraffins

Answer: Option A — Paraffins, Olefins, and Acetylenes

57. Isomers that differ in partial arrangement of atoms rather than order of atomic connectivity are called ___ (A) Constitutional isomers (B) Skeletal isomers (C) Position isomers (D) Stereoisomers

Answer: Option D — Stereoisomers

58. Pairs of optical isomers whose mirror images are not superimposed are called _____ (A) Diastereomers (B) Isomerism (C) Enantiomers (D) Stereoisomers

Answer: Option C — Enantiomers

59. A 50:50 mixture of enantiomers with no observable optical activity are called ____ (A) Pure enantiomer (B) Racemic mixture (C) Diastereomer (D) Meso compound

Answer: Option B — Racemic mixture or racemate

60. Two diastereomers that differ only at one stereo center are referred to _____ (A) Enantiomers (B) Epimers (C) Anomers (D) Regioners

Answer: Option B — Epimers

61. A device through which only light waves oscillating in a single plane can pass is called ___ (A) Polarizer (B) Polarimeter (C) Both (D) None

Answer: Option A — Polarizer

62. Which of the following groups has the highest priority based on sequence rules in the specification of R or S configuration in stereoisomers? (A) –SO■H (B) –OH (C) –CH=CH■ (D) –CH■CH■

Answer: Option B — –OH

63. Two of these are typical properties of transition metals (A) Brittle and hardness (B) High melting points and lustre (C) Variable oxidation state and colour (D) Hardness and ductility

Answer: Option C — Variable oxidation state and colour

64. The inner transition elements are made of two series (A) The lanthanides & actinides series (B) The 3d & 4d transition metals series (C) The 4d & 5d transition metal series (D) The radioactive & non-radioactive series

Answer: Option A — the lanthanides and actinides series

65. Which of these statements is incorrect? (A) Crystalline solids possess long range order (B) There is a regular arrangement of atoms in crystalline solids (C) Amorphous solid has a short range order (D) There is a periodic arrangement of atoms in amorphous solids.

Answer: Option D — There is a periodic arrangement of atoms in amorphous solids

66. _____ is defined as the smallest group that repeats itself over and over again to give the full symmetry of the crystal ? (A) Unit cell (B) Point lattice (C) Crystal system (D) Motif

Answer: Option A — Unit cell

67. What is the IUPAC name for CH3CH(OH)CH2COCH2COOH? (A) 2-hydroxy-4-oxohexanoic acid (B) 5-hydroxy-3-oxohexanoic acid (C) 6-carboxy-3-oxohexanoic acid (D) 6-carboxy-5-hydroxyhexan-3-one acid

Answer: Option B — 5-hydroxy-3-oxohexanoic acid

68. What is the common name for benzenol whose second hydroxyl group is at the Meta position?* (A) Hydroquinone (B) Catechol (C) Cresol (D) Resorcinol

Answer: Option D — Resorcinol

69. The reagent that can be used to distinguish the extent of acidity of phenol and carboxylic acid is (A) Tollen reagent (B) Lucas reagent (C) Fehling reagent (D) Sodium hydrogen trioxocarbonate reagent

Answer: Option D — Sodium hydrogen trioxocarbonate regent

70. Aluminium and gallium form oxide and hydroxide that are _____ (A) Non-Acidic (B) Acidic (C) Basic (D) Amphoteric

Answer: Option D — Amphoteric.

71. The given reaction, $2CuCl \rightarrow Cu + CuCl_2$ is an example of which type of reaction? (A) Precipitation (B) Decomposition (C) Synthesis (D) Disproportionation

Answer: Option D — Disproportionation

72. _____ is the allotrope carbon that conducts electricity (A) Diamond (B) Fullerenes (C) Graphite (D) Charcoal

Answer: Option C — Graphite

73. The ability of carbon atoms to self-link in forming long chains and ring compounds is known as (A) Photolysis (B) Hydrolysis (C) Synthesis (D) Catenation

Answer: Option D — Catenation

74. The process of mixing orbitals of different energies to give a set of orbitals of equivalent energy is called (A) Orbital mixing (B) Equivalent energy (C) Specialization (D) Hybridization Cl |

Answer: Option D — Hybridization

75. H3CCH —C —CH2 — CH3 | i.e. 2-Chloro-2-methylbutane H3C What type of hybridization is found in the above compound? (A) Sp (B) Sp^2 (C) Sp^3 (D) None

Answer: Option C — Sp^3

76. In the hybridization of atomic orbitals of carbon, what is the bond angle between orbitals in sp^2 and sp^3 hybridization respectively? (A) 109.5° and 180° (B) 109.5° and 120° (C) 180° and 109.5° (D) 120° and 109.5°

Answer: Option D — 120° and 109.5°

77. Carbon is able to form numerous compounds because of its ability to (A) Donate electrons (B) Gain electrons (C) Catenate (D) Form ionic bond

Answer: Option C — Catenate

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1. Which of the following compounds is formed when bromine water reacts with unsaturated hydrocarbons? (A) Hydroxybromine (B) Bromoalkyne (C) Anhydrous hydrogen bromide (D) Hydroxy hydrogen bromide

Answer: Option B — Bromoalkyne

2. The IUPAC name for CH■CH(CH■)CH=CHCH■CH(OH)CH■COCH■ is (A) 4-Hydroxy-8-methylnon-6-en-2-one (B) 2-Methyl-6-hydroxynon-5-en-2-one (C) 2-Methyl-8-oxonon-4-en-6-nol (D) 4-Hydroxy-2-oxo-2-methylnon-4-ene

Answer: Option A — 4-Hydroxy-8-methylnon-6-en-2-one

3. Methanol reacts with sodium metal to form: (A) Sodium ethoxide and oxygen (B) Sodium methoxide and oxygen (C) Sodium ethoxide and hydrogen (D) Sodium methoxide and hydrogen

Answer: Option D — Sodium methoxide and hydrogen

4. Lucas reagent used to distinguish the three classes of alcohol is a mixture of (A) MgCl■/H■SO■ (B) ZnCl■/H■SO■ (C) MgCl■/HCl (D) ZnCl■/HCl

Answer: Option D — ZnCl■/HCl.

5. When a keto group is directly attached to a benzene ring, the compound formed is known as (A) Toluene (B) Benzonol (C) Phenone (D) Benzoic acid

Answer: Option C — Phenone

6. Which of the following compounds is used as an anti-freezing agent (A) Glycerol (B) Ethylene glycol (C) Mannitol (D) Acetylene

Answer: Option B — Ethylene glycol

7. Which of the following is a secondary amine (A) CH■CH■NH■ (B) CH■N(CH■)■ (C) (CH■)■NH (D) (CH■)■NH2HCl

Answer: Option C — (CH■)■NH

8. Functional group in an unknown sample can be identified by employing (A) Fourier Transform Infrared Spectroscopy (B) Mass spectroscopy (C) Ultraviolet spectroscopy (D) X-ray crystallography

Answer: Option A — Fourier Transform Infrared Spectroscopy (FTIR)

9. What are responsible for the coloration of organic compounds (A) Functional groups (B) Chromophores (C) Hybridization (D) Isomerism

Answer: Option B — Chromophores.

10. A yellow non-sooty flame is indicative of the presence of (A) Unsaturated Aliphatic compounds (B) Saturated Aliphatic compounds (C) Unsaturated Aromatic compounds (D) Saturated Aromatic compounds

Answer: Option B — Saturated aliphatic.

11. Halogen can be detected in the organic residue by (A) Carus test (B) Duma method (C) Fajan method (D) Beilstein test

Answer: Option D — Beilstein test.

12. One of these is commonly used to dissolve the ignition residue for further analysis (A) Concentrated HCl (B) Concentrated H■SO■ (C) Concentrated HNO■ (D) CH■COOH

Answer: Option A — Conc. HCl.

13. Residue after ignition of an unknown organic compound is suggestive of (A) The presence of double bond (B) The presence of a metal (C) The presence of single bonds (D) The presence of sulphur

Answer: Option B — the presence of a metal

14. Quantitative analysis for nitrogen is carried out by (A) Kjeldahl's method (B) Volhard's method (C) Mohr's method (D) The electrolytic method

Answer: Option A — Kjeldahl's method

15. The single bond between the carbon atoms causes methane, CH■, to have which of the following shapes? (A) Trigonal planar (B) Linear (C) Tetrahedral (D) Trigonal bipyramidal

Answer: Option C — Tetrahedral.

16. What prefix is used when 2 carbon atoms are removed from the parent alkane compound? (A) Seco (B) Bisnor (C) Des (D) Cyclo

Answer: Option B — Bisnor.

17. ___ number of isomers can be possibly obtained from octane (A) 78 (B) 18 (C) 35 (D) 25

Answer: Option B — 18

18. Which hydrocarbon would be expected from the reaction of n-butyl chloride with sodium metal? (A) Octane (B) Decane (C) Nonane (D) Pentane

Answer: Option A — Octane .

19. Provide the correct IUPAC name for 1,3-dimethyl butane (A) 1-methyl pentane (B) 2-methyl butane (C) 1-methyl butane (D) 2-methyl pentane

Answer: Option D — 2-methyl pentane.

20. A soft silvery metal with a melting point of 98°C that belongs to group one is: (A) Sodium (B) Potassium (C) Lithium (D) Boron

Answer: Option B — Potassium.

21. A white crystalline solid with a M.P of 801°C that conducts electricity either in the molten state or aqueous solution is: (A) KCl (B) ZnCl■ (C) NaCl (D) MgCl■

Answer: Option C — NaCl.

22. Identify the correct electronic configuration for Chloride ion. (A) $1s^2 2s^2 2p■ 3s^2 3p■$ (B) $1s^2 2s^2 2p■ 3s^2 3p■$ (C) $1s^2 2s^2 2p■ 3s^2 3p■$ (D) $1s^2 2s^2 2p■ 3s^2 3p^3$

Answer: Option C — $1s^2 2s^2 2p■ 3s^2 3p■$

23. According to Hess's law, (A) ∆H = ∆H■ + ∆H■ (B) ∆H = ∆H■ x ∆H■ (C) ∆H = ∆H■ + ∆H■ + ∆H■ (D) ∆H = ∆H■ x ∆H■ x ∆H■

Answer: Option C — ∆H = ∆H■ + ∆H■ + ∆H■.

24. For any stable compound to be formed from its elements, there must be a net ___ of its potential energy. (A) Increase (B) Lowering (C) Stabilization (D) Removal

Answer: Option B — Lowering

25. Which of the following carboxylic acids has the highest boiling point? (A) Octanoic acid (B) Heptanoic acid (C) Decanoic acid (D) Hexanoic acid

Answer: Option C — Decanoic acid

26. The IUPAC name of COOH | COOH (A) Propanedioic acid (B) Butanedioic acid (C) Ethanoic acid (D) Ethanedioc acid

Answer: Option B — Ethanedioc acid

27. Lower carboxylic acids are soluble in water due to the presence of (A) Covalent bonds (B) Hydrogen bonds (C) None of the above (D) All of the above

Answer: Option B — Hydrogen bonds.

28. Which of the following reactions of carboxylic acids serves as a confirmatory test for its presence? (A) Reaction with litmus paper (B) Reaction with trioxocarbonates (C) Ester formation (D) Amide formation

Answer: Option B — Reaction with trioxocarbonates

29. The predicted bond angle in methane molecule is__ (A) 109.5° (B) 107.5° (C) 120° (D) 180°

Answer: Option A — 109.5° OR 109°28′.

30. Which one of the following is a limitation of the lewis model? (A) This model could not give the bond angles and shapes of molecules (B) It explains the shapes and bonds of angles of molecules (C) It is used to show shared electron pairs (D) It is used to show shared ion pairs of electrons

Answer: Option A — This model could not give the bond angles and shapes of molecules.

31. The following are not types of solids except, (A) Crystalline and amorphous solids (B) Amorphous and coordinated solids (C) Irregular and crystalline solids (D) Conjugated solids

Answer: Option D — Conjugated solids.

32. Anisotropy is to __ solids as isotropy is to __ solids (A) Regular and finite solids (B) True solids and pseudo solids (C) Amorphous and coordinated solids (D) Conjugated and coordinated solids

Answer: Option D and C respectively — Anisotropy is to conjugated and coordinated solids, as isotropy is to amorphous and coordinated solids.

33. What is the binding force in metallic solids? (A) Electrostatic attraction (B) Electrostatic attraction between positive and negative ions (C) Ionic interaction (D) Van der Waals interaction

Answer: Option A — Electrostatic attraction.

34. CH■ is ___ in shape. (A) Trigonal pyramidal (B) Angular (C) Tetrahedral (D) Trigonal

Answer: Option C — Tetrahedral in shape.

35. According to valence bond theory, a covalent bond is formed by overlapping of _____ (A) Two half-filled atomic orbitals containing electrons of opposite spins (B) Two half-filled atomic orbitals containing the same spins (C) Four half-filled atomic orbitals (D) Two filled atomic orbitals with two electron spins

Answer: Option A — Two half-filled atomic orbitals containing electrons of opposite spins.

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1. The chemistry of the elements of group one is dominated by ___ except (A) Electron affinity (B) Ionization energy (C) Electrospositivity (D) Electronegativity

Answer: Option C — electropositivity.

2. Members of group one present as ___ during redox agents (a) Redox agent (b) Catalytic agent (c) Reducing agent (d) Spectral dichroism

Answer: Option C — Reducing agent.

3. Which of the following is not involved in the reaction 2Na + Cl■ → 2NaCl (A) Atomization (B) Ionization (C) Electron affinity (D) Hydration

Answer: Option D — Hydration

4. What is the gas that shall be evolved when sodium metal reacts with cold water? (A) Chlorine gas (B) Hydrogen gas (C) Oxygen gas (D) Water vapour

Answer: Option B — Hydrogen gas

5. The characteristic feature of transition metals is the possession of (A) s–orbital (B) p–orbital (C) d–orbital (D) quantum orbital .

Answer: Option C — d-orbital

6. Which of these is a transition metal (A) H (B) Na (C) Ca (D) Fe

Answer: Option D — Fe

7. An organic compound which liberates carbon (iv) oxide from trioxocarbonate (iv) solution is likely to be (A) CH■OH (B) C■H■ (C) C■H■ (D) CH■COOH

Answer: Option D — CH■COOH

8. The relatively high boiling points of carboxylic acid are due to (A) Hydrogen bonding (B) Covalent bonding (C) Atomic character (D) Ionic bonding

Answer: Option A — Hydrogen bonding.

9. Cyclic esters called (A) Lactones (B) Alkanones (C) Quinones (D) Ketones

Answer: Option A — Lactones

10. Another name for ethanoic acid is _____ (A) Acetic acid (B) Maleic acid (C) Propanoic acid (D) Succinic acid

Answer: Option A — Acetic acid

11. Chemical identification of ester can be achieved by the use of _____ (A) Lucas reagent (B) Toluene reagent (C) Fehling's test (D) Hydroxamic acid test

Answer: Option D — Hydroxamic acid test

12. What is the IUPAC name of CH■CH■CHO (A) Propanol (B) Propanal (C) Propanone (D) Propanoic acid

Answer: Option B — Propanal

13. ___ gas is liberated when ethanol reacts with sodium metal (A) H■ (B) O■ (C) N■ (D) Water vapour

Answer: Option A — H■

14. What is the common name for methyl benzene (A) Cresol (B) Toluidine (C) Toluene (D) Pitenone

Answer: Option C — Toluene

15. When a ketone group is directly attached to an aromatic ring, the compound is called (A) Toluene (B) Aniline (C) Phenone (D) Toluidine

Answer: Option C — Phenone

16. _____ reagent is used to differentiate aldehydes from ketone.. (A) Fehling's solution (B) Tollen's reagent (C) 2,4-dinitrophenylhydrazine (D) Benedict's reagent

Answer: Option A — Fehling's solution.

17. _____ reagent is used to distinguish saturated from unsaturated hydrocarbons. (a) Fehling's solution (b) Tollen's reagent (C) Benedict reagent (d) Bromine water

Answer: Option D — Bromine water.

18. Ethanol is an example of _____ (a) Primary alcohol (b) Secondary alcohol (c) Tertiary alcohol (d) Quaternary alcohol

Answer: Option A — Primary alcohol

19. The hydrocarbon in which all the 4 valencies of carbon are fully occupied is called _____ (a) Alkenes (b) Alkynes (c) Alkanes (d) Cycloalkane

Answer: Option C — Alkane.

20. Chlorination of alkanes is an example of free radical ___ reaction (a) Rearrangement (b) Elimination (c) Substitution (d) Addition

Answer: Option C — Substitution reaction

21. What prefix is used when 2 carbon atoms are removed from the parent alkane compound? (a) Seco (b) Bisnor (c) Des (d) Cyclo

Answer: Option B — Bisnor

22. Which hydrocarbon would be expected from the reaction of n-pentyl chloride with sodium metal? (a) Octane (b) Decane (c) Nonane (d) Pentane

Answer: Option B — Decane

23. One of these is not commonly employed in qualitative analysis (a) NaOH (b) HCl (c) HNO3 (d) Ascorbic acid

Answer: Option D — Ascorbic acid

24. What is the colour of the precipitate obtained in the test for sulphur? (a) White (b) Blue (c) Black (d) Yellow

Answer: Option C — Black.

25. One of these can replace Na in Lassaigne's test (a) K (b) Li (c) Mg (d) Ca *ANSWER:* This question is on color; *no complete

Answer: .

26. A yellow sooty flame indicates _____ (A) Presence of oxygen (B) Aromaticity and unsaturation (C) High carbon content (D) None

Answer: Option B — Aromaticity and unsaturation.

27. Preliminary examination of specimen has to do with all except (A) Physical state (B) Color (C) Odor (D) Solubility

Answer: Option D — Solubility

28. X is a white precipitate and soluble in ammonia hydroxide, what is X? (A) Sulphur (B) Chlorine (C) Iodine (D) Fluorine

Answer: Option B — Chlorine

29. The basic building block of crystal is (A) Unit cell (B) Lattice (C) Formulae unit (D) Atom

Answer: Option A — Unit cell

30. Which of the following is not a class of solid on the basis of bonding type? (A) Metallic solids (B) Ionic solids (C) Covalent solids (D) Semi-conductors

Answer: Option D — Semiconductors

31. One of this information cannot be obtained from a balanced chemical equation (A) State of the reactants/products (B) Direction of reaction (C) Speed of reaction (D) Product formed

Answer: Option C — Speed of reaction

32. The formation of stable ionic solids is favoured by the following factors except (a) Low ionization energy (b) High electron affinity (c) High lattice energy (d) Low solvation energy

Answer: Option D — Low solvation energy.

33. Amorphous solids have ____ structure (a) Regular (b) Linear (c) Irregular (d) Dentrilic

Answer: Option C — Irregular.

34. The major type of bonding that exist between organic molecules is ____ (a) Covalent bond (b) Ionic bond (c) Dative bond (d) Co-ionic bond

Answer: Option A — Covalent bond.

35. NH3 has ____ lone pair(s) of electrons (a) 1 (b) 2 (c) 3 (d) 4

Answer: Option A — 1 SECTION B: TEST OF LABORATORY EXPERIENCE

Section B — Test of Laboratory Experience

1. brownish colored organic liquid specimen can be said to contain ___ functional group .

Answer: Amines

2. A specimen of ester will emit ___ smell or odour.

Answer: Esters tend to have SWEET/pleasant odour

3. An organic specimen containing ___ will combust in the non-luminous (blue) zone of a Bunsen burner flame to leave white residue.

Answer: Metallic Salt

4. A pure organic specimen shall show a ___ melting point.

Answer: Sharp Melting Point

5. __ and __ apparatus can be used in the laboratory to determine the melting point of the given organic specimen.

Answer: Melting points* are routinely measured in many laboratories using the following apparatus: Thiele tube (25°C - 180°C) Thomas Hoover (25°C - 300°C) Fisher-Johns (25°C - 300°C) Mel-Temp (25°C - 400°C)

6. The process of reclaiming a specimen from its solution by heating to remove dryness or partial dryness and kept to cool is called ___

Answer: Evaporation to dryness*

7. A pure sample can be obtained from its crude by ___ process.

Answer: Chromatography, Recrystallization, Sublimation or Distillation process.

8. __ and __ are among the solvents that may be used to test the solubility of benzoic acid.

Answer: Ethanol, Propanone(Acetone), Tetrachloro methane and Water. CHM 102

CHM 102 — 2021/2022 Set

1. The modification of the periodic law was as a result of the discovery of __ by __ (A) Atomic number by Henry Moseley (B) Atomic mass by Isaac Newton (C) Atomic number by Ernest Rutherford (D) Atomic number by James Chadwick

Answer: Option A — Atomic number by Henry Moseley

2. Which of the following is an alkane with optical activity? (A) 3-methylhexane (B) Isopentane (C) Neopentane (D) 3-methylpentane

Answer: Option A — 3-methylhexane

3. Octave law was used as basis for classification of elements by (A) Moseley (B) Dobereiner (C) Mendeleev (D) Newland

Answer: Option D — Newland

4. Prefix indicates rings removal by microorganism and/or thermal stress (A) Sec (B) Bisnor (C) Des (D) Bispir

Answer: Option C — Des

5. A yellow non-sooty flame is associated with (A) Double bond (B) Nitrogen (C) Respiration (D) Aliphatic chains

Answer: Option D — Aliphatic chain

6. Residue after combustion indicates the presence of (A) Double bond (B) Saturation (C) Metal (D) Impurity

Answer: Option C — Metal

7. CH■CH■CO■H + C■H■OH ■ X + H■O, X in the reaction is (A) Ethylpropanoate (B) Methylmethanoate (C) Pentanoic acid (D) Propylethanoate

Answer: Option A — Ethylpropanoate

8. Cyclic esters are called (A) Quinone (B) Lactone (C) Ketone (D) Alkane

Answer: Option B — Lactone

9. CH■CH■CO■H + C■H■OH ■ X + H■O. The structural formula of X in the reaction above is _____ A. CH■CH■COOC■H■ B. CH■COOCH■CH■CH■ C. CH■CH■CH■CH■COOH D. CH■COOC■H■

Answer: Option A — CH■CH■COOC■H■ i.e. Ethylpropanoate.

10. The outermost electronic configuration, ns^2np^2 is that of A. None of the above B. Group IV C. All p-block elements D. Transition elements

Answer: Option B — Group IV

11. Examples of metals that crystallize in face-centered cubic lattice are A. Fe and Na B. Mg and Zn C. Al and Cu D. Al and Au

Answer: Option C — Al and Cu

12. The smallest formula that shows the relative number of different kinds of atoms in a molecule A. Molecular formula B. Molar mass C. Empirical formula D. Formula weights

Answer: Option C — Empirical formula.

13. CH■CH■CO■H + C■H■OH ■ X + H■O. The reaction above is called A. Halogenation B. Amide formation C. Halide formation D. Esterification

Answer: Option D — Esterification and the product, X formed is Ethylmethanoate.

14. The d-block elements constitute A. Alkali metals B. Transition metals C. Alkaline earth metals D. Chalcogens

Answer: Option B — Transition metals

15. Thermal and electrical properties distinguish A. Metals and ions B. All of the above C. Metals and non-metals D. Non-metals and ions

Answer: Option C — Metals and non-metals

16. What is the molecular formula for the alkane of smallest molecular weight which possesses a stereogenic centre A. C■H■■ B. C■H■■ C. C■H■ D. C■H■■

Answer: Option A — C■H■■

17. What is the IUPAC name for CH■CH(CH■)CH=CHCOCH■ A. 5-Methylhept-3-en-2-one B. 5-Methylhex-3-en-2-one C. 2-Methylhept-3-en-2-one D. 2-Methylhex-3-en-2-one

Answer: Option B — 5-Methylhex-3-en-2-one

18. Give the group VII electronic configuration A. ns■np² B. ns²np■ C. ns■np■ D. ns²np■

Answer: Option D — ns²np■

19. How many stereoisomers is possible for 2,3-pentanediol? A. 6 B. 4 C. 2 D. 5

Answer: Option B — 4

20. What prefix is used when 2 carbon atoms are removed from the parent alkane compound? A. Bisnor B. Des C. Sec D. Cyclo

Answer: Option A — Bisnor

21. The boiling point and freezing point of carboxylic acids are usually much higher than those of its isomeric compounds due to the presence of – A. Metallic bond B. Hydrogen bond C. Ionic bond D. Van der Waals forces

Answer: Option B — Hydrogen bonding

22. Which of the following reagent is used for the identification of terminal alkynes A. Tollen's reagent B. NH■OH/AgNO■ C. NH■OH/CuCl D. NH■OH/CuCl■

Answer: Option C — NH■OH/CuCl

23. Which of the following alkynes can be identified and distinguished from the rest of the alkynes on reaction with ammoniacal silver nitrate to give a white precipitate?* A. CH■C≡C–CH■ B. CH■CH■C≡CH C. CH■CH■C≡CCH■ D. CH■C≡CCH■CH■CH■

Answer: Option B — CH■CH■C≡CH

24. Which of the following compounds will be the most soluble in water? A. Pentane B. Pentanoic acid C. Ethanoic acid D. Ethane

Answer: Option C — Ethanoic acid CH■COOH

25. Which of these pairs is fully s-block? A. Na and Ca B. Na and C C. Li and Al D. Li and N

Answer: Option A — Na and Ca

26. S-block elements A. Have none of their electrons in an s-orbital B. Have all their electrons in s-orbital C. Have most of their electrons in an s-orbital D. Have their valence electrons in an s-orbital

Answer: Option D — Have their valence electrons in an s-orbital

27. Aldehyde are said to be reducing agent due to?. A. Presence of single hydrogen atom attached to the carbonyl carbon B. Absence of single hydrogen atom attached to the carbonyl carbon C. Oxidizing power D. Acid formation

Answer: Option A — presence of a single hydrogen atom attached to the carbonyl carbon.

28. Quantitative analysis for nitrogen can be executed by A. Nitration test B. Neutralization C. Kjeldahl method D. Solubility test

Answer: Option C — Kjeldahl method

29. CH■COOH + A ■ CH■CONH■ + H■O is A. Amide B. Amine C. Ammonium hydroxide D. Ammonia

Answer: Option B — Amine

30. Compounds which have different arrangement of atoms in space while having same atoms bonded to each other are said to be A. Positional isomerism B. Chain isomerism C. Stereoisomerism D. Functional group isomerism

Answer: Option C — Stereoisomerism.

31. Which of the hydrocarbon would be expected from the reaction of n-pentyl chloride with sodium metal? A. Nonane B. Pentane C. Decane D. Octane

Answer: Option C — Decane CHM 102

CHM 102 — 2019/2020 Examination

1. The arrangement of particle in a crystalline solids are determined indirectly by (A) X-ray diffraction (B) FTIR (C) UV-vis spectroscopy (D) Raman spectroscopy

Answer: Option A — X-ray diffraction

2. The smallest volume is a crystal that shows all the characteristics of the crystal's pattern is called ____ (a) unit cell (b) crystal structure (c) crystal lattice (d) lattice point

Answer: Option A — unit cell

3. An example of molecular solid is _____ (a) Li (b) K■SO■ (c) P■ (d) C (diamond)

Answer: Option C — P■

4. Nearly all metals crystallize in one of three types of lattices except_____ (a) Body centered cubic, BCC (b) Face-centered cubic, FCC (c) Hexagonal close-packed HCP (d) Extended rigid crystalline network.

Answer: Option D — Extended rigid crystalline network

5. Most salts crystallize as ionic salts with ___ occupying the unit cells. (A) ions (B) molecules (C) atoms (D) electrons

Answer: Option A — ions

6. A band within which (or into which) electrons must move to allow electrical conduction is called (A) conduction band (B) band gap (C) forbidden zone (D) small energy gap

Answer: Option A — Conduction Band

7. Crystalline non-metals, such as Diamond and Phosphorous, are insulators (A) They don't conduct electricity (B) They are malleable and/or ductile (C) They are brittle (D) They are too hard

Answer: Option A — They don't conduct electricity

8. The doctrine of vitalism believes that living systems possessed ___ (A) mean force (B) vital force (C) eternal energy (D) vital power

Answer: Option B — Vital Force

9. Compound with the same elemental composition but different structural arrangement are called ___ (A) Monomer (B) Isomers (C) Polymers (D) Allotropy's

Answer: Option B —Isomers

10. The boiling point of carboxylic acids are higher than expected due to ___ (A) hydrogen bond (B) Sigma bond (C) ionic bond (D) Covalent bond

Answer: Option A — hydrogen bond

11. Which of the following reactions of carboxylic acids serves as a confirmatory test for their presence? (A) Reaction with litmus paper (B) Reaction with trioxocarbonates (C) Ester formation (D) Amide formation

Answer: Option A — Reaction with litmus paper

12. One of these is a chemical reaction of carboxylic acid (A) Amide formation (B) Esterification (C) Anhydride formation (D) All of the above

Answer: Option D — All of the above

13. Acid reactions of the carboxylic acid includes the following except (A) Changing of blue litmus paper to red (B) Reacting with NaOH(aq) (C) Reaction with ethanol (D) Reaction with Carbonates

Answer: Answer — All of the above

14. Almost all the elements of the s-block in the periodic table show ___ property (A) Radioactive (B) Affinity (C) Metallic (D) Reactive

Answer: Option C — Metallic

15. Members of group 3 or 13 in the periodic table can be referred to as _____ (A) Metals (B) Non-metals (C) Metalloids (D) d-block metals

Answer: Option A — Metals

16. ___ bonds hold the particles of metals in their free state together. (A) Cohesion (B) Van der Waals (C) Ionics (D) Metallic

Answer: Option D — Metallic

17. The most probable reaction between members of group one (1) and seventeen (17) is ___ (A) Atomic (B) Nuclear (C) Electron (D) Ionic

Answer: Option D — Ionic bonding

18. Salts can conduct electricity in all state except ___ (A) Solids (B) Molten (C) Dissolved (D) Ionic

Answer: Option A — Solids

19. _____ among the following is not a metal (A) Na (B) Hg (C) Pb (D) None

Answer: Option D — None

20. Metals show all of the following properties except__ (A) Opaque (B) Dense (C) Cationic (D) Covalence

Answer: Option D — Covalence

21. _____ prefix indicates rings removal by microorganism and/or thermal stress (A) Nor (B) Bisnor (C) Des (D) Seco

Answer: Option C — Des

22. How many isomers can be obtained from nonane? (A) 78 (B) 35 (C) 18 (D) 9

Answer: Option B — 35

23. In Wurtz synthesis, when an alkyl halide such as chloroethane reacts with sodium metal, the type of alkane formed under an ether catalyst is (A) Pentane (B) Ethane (C) Propane (D) Butane

Answer: Option D — Butane

24. As compared to its parent alkanes, an alkyl radical contains (A) One less carbon (B) One less hydrogen (C) One more carbon (D) One more hydrogen

Answer: Option B — One less hydrogen

25. What is the correct IUPAC nomenclature in the picture below? (A) 2-methyl Pentane (B) 1,2-dimethyl Pentane (C) 2-methyl Butane (D) 2-ethyl heptane

Answer: Option A — 2-methyl Pentane

26. The presence of halogen can be detected by ___ test. (A) Beilstein's test (B) Millions test (C) Burette test (D) Filtration

Answer: Option A — Beilsteins

27. The residue after combustion is an indicative of ___ (A) Metals (B) Nonmetals (C) Oxides (D) Isomers

Answer: Option A — Metals

28. Quantitative analysis of nitrogen is carried out by which analysis? (A) Kjeldahl method (B) Dumas method (C) Titration (D) Nitratolab

Answer: Option A — Kjeldahl method

29. A certain hydrocarbon contained 80% of carbon by mass. If the relative molecular mass is 30g/mol. Determine its relative molecular formula. (A) CH■ (B) C■H■ (C) C■H■ (D) C■H■

Answer: Option C — C■H■

30. One of these will exhibit optical activity: (A) Neopentane (B) Isopentane (C) 3-methylpentane (D) 3-methylhexane

Answer: Option D — 3-methylhexane

31. One of these is a meso compound (A) 2,3-Dichloropentane (B) 2,4-Dichloropentane (C) 1,3-Dimethylcyclopentane (D) 5-Aminobenzenoic acid

Answer: Option B — 2,4-Dichloropentane

32. Keto-enol is an example of — (A) Tautomers (B) Enantiomers (C) Diastereomers (D) Racemate

Answer: Option A — Tautomers

33. A compound with triple bond is most likely to have one of these geometry (A) Tetrahedral (B) Trigonal (C) Linear (D) Square planar

Answer: Option C — Linear

34. Non-crystalline solid having no well-defined and ordered structure is referred to as (A) Amorphous (B) Crystalline (C) Random (D) Structured

Answer: Option A — Amorphous

35. Immiscible liquids can be separated through (A) Centrifugation (B) Decanting (C) Separating funnel (D) Evaporation

Answer: Option C —Separating funnel Wishing you Success in your exams ■